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Education

Shiraz University

Fall 2023 – Summer 2027

B.Sc. in Chemical Engineering

Shiraz, Iran

- **Cumulative GPA: 17.90 / 20.00** (last two years: **18.47 / 20.00**, estimated). Ranked **2nd** in class to date; consistently among the top three of the cohort by semester.
- Perfect **20.00 / 20.00** in six core courses: Mass Transfer, Unit Operations I, the Heat Transfer and Fluid Mechanics laboratories, Numerical Methods, and Introduction to Biological Sciences.

Notable Coursework

Transport & Separations	Fluid Mechanics II (18.90), Heat Transfer I (18.90), Combustion (18.00), Heat Transfer II (16.55)
Thermodynamics & Kinetics	Thermodynamics I (19.50), Thermodynamics II (19.30), Chemical Kinetics and Reactor Design (16.96)
Computation & Mathematics	Engineering Mathematics (19.40), Computer Programming (19.00)
Chemistry	General Chemistry II (19.50), Organic Chemistry (19.30), Physical Chemistry (18.83), Analytical Chemistry (18.75)

Research Experience

Sturgeon Research Facility

Summer 2026 – Present

Biochemistry Laboratory Intern

Rasht, Iran

- Perform biochemical assays on tissue samples, including **catalase activity** and **malondialdehyde (MDA) quantification**, following standardised assay protocols.
- Prepare tissue homogenates, buffers, reagents, and calibration standards by micropipetting; process samples through **homogenisation and centrifugation**.
- Measure absorbance by **spectrophotometry** and determine biochemical quantities from experimentally generated **standard curves**.

Research & Course Projects

Computational McCabe–Thiele Distillation Simulator

MATLAB, JavaScript

Self-directed project, Unit Operations

- Developed a modular implementation of the **McCabe–Thiele method** for multi-section distillation columns with arbitrary feeds, side streams, and subcooled reflux.
- Implemented section-wise material balances, q -lines and operating lines, vapour–liquid equilibrium fitting and interpolation, curve-intersection detection, Murphree efficiency with pseudo-equilibrium correction, and automated stage stepping.
- Structured the solver as reusable functions behind a single execution script so it applies across a range of Unit Operations problems, then ported the validated model to a browser-based **JavaScript/HTML** implementation.

Coupled Transport Modelling of Evaporative Cooling

COMSOL Multiphysics

Mass Transfer course project

- Built a **three-dimensional COMSOL model** coupling turbulent airflow, heat transfer, and evaporative mass transfer for cooling of a liquid-filled ceramic vessel.
- Solved a stationary turbulent-flow field followed by transient heat and moisture transport, then investigated the effects of **ambient temperature, relative humidity, and latent heat of vaporisation**.
- Quantified the contribution of evaporation: neglecting latent heat shifted the predicted 10-minute liquid temperature by roughly **14%**, while relative humidity produced a substantially smaller effect.

Nonlinear Material-Balance Modelling of Crystallisation

MATLAB

Numerical Methods course project

Grade: 20 / 20

- Formulated a **six-variable nonlinear material-balance model** for purification by crystallisation and implemented a Newton-based solution scheme.
- Investigated the convergence limitations of the nonlinear system and, once the Newton formulation failed to converge, recovered candidate solutions through symbolic solution instead.

- Validated the result independently by reformulating the balances around different process boundaries and running component-level conservation checks.

Biological Sorption: Biosorption and Bioaccumulation

Literature review

Introduction to Biological Sciences

Grade: 20 / 20

- Surveyed the literature on **biosorption and bioaccumulation**, connecting biological sorption mechanisms to the adsorption and separation framework used in chemical engineering.
- Examined adsorption equilibria, isotherm modelling, batch material balances, and the roles of equilibrium and kinetics in adsorption dynamics.

Open-Source Scientific Software

cheme-thermo-solvers: Phase Equilibrium Library

MATLAB

Independent open-source project

 [GitHub](#)

- Wrote a suite of iterative solvers for multicomponent **vapour–liquid equilibrium**, bubble- and dew-point calculations, and non-ideal liquid-phase activity via the **UNIFAC group-contribution method**.

numerical-methods-cheme: Regression Engine

MATLAB

Independent open-source project

 [GitHub](#)

- Built **LCInterpol**, a multi-variable nonlinear regression engine, and applied it to parameter estimation for **Arrhenius** rate expressions and growth-kinetics models.

Teaching & Outreach

Shiraz University

Fall 2025

Teaching Assistant, Thermodynamics I

Shiraz, Iran

- Led problem-solving sessions on assigned homework and supplementary examination-preparation problems, explaining solution strategies and core thermodynamic concepts to support student understanding.

Student Learning Centre for Accessible Learning

Volunteer Peer Instructor

Shiraz, Iran

- Taught material balances, thermodynamics, and fluid mechanics to groups of petroleum engineering students.

Free Discussions in Foreign Languages

Summer 2024

Coordinator & Session Leader

Shiraz, Iran

- Coordinated and led **12 free discussion sessions** in English and German, open to all participants.

Technical Skills

Multiphysics & CFD	COMSOL Multiphysics: turbulent flow, heat transfer, transient moisture transport, parametric studies
Programming & Numerics	MATLAB (nonlinear systems, symbolic computation, regression and curve fitting, interpolation); JavaScript/HTML; $\text{\LaTeX}$
Process Engineering	McCabe–Thiele distillation analysis, vapour–liquid equilibrium and UNIFAC, crystallisation material balances, transport-phenomena modelling
Biochemical Laboratory	Tissue homogenisation, micropipetting, buffer and standard preparation, centrifugation, spectrophotometry, catalase and MDA assays, standard-curve quantification
Research & Communication	Literature review, technical report writing, scientific presentation, experimental data interpretation, technical documentation

Honours & Awards

Top 1% of 136,000 participants, Iranian National University Entrance Examination (*Konkur*)

2023

Qualified past the first round, Iranian National Chemistry Olympiad

Summer 2021

Certifications, Languages & Professional Development

- **Health, Safety and Environment (HSE) Certificate**, Shiraz University.
- **Languages:** Persian (native) · English (advanced/fluent) · German (intermediate, A2–B1).
- **Seminar on floating solar photovoltaic systems**, Shiraz University, Fall 2023; attended alongside Ministry of Energy representatives.